Predictive Models for Student Dropout Risk:
A Review and Empirical Evaluation Using Real-World Data

A Project Submitted in Partial Fulfillment of the Requirements
for the Degree of Bachelor of Science (BSc) in
Statistics and Statistics and Computer Science

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ABSTRACT

This study focuses on the problem of student dropout in higher education and explores the use of predictive models as early warning systems to identify students at risk of dropping out. Student dropout represents a major challenge for educational institutions, particularly in low-resource and conflict-affected environments such as Sudan, where academic disruption and limited institutional support increase dropout rates.

The study applies an applied predictive approach using the Open University Learning Analytics Dataset (OULAD), which contains demographic, academic, and behavioral data related to students' engagement with virtual learning environments. Several machine learning and statistical models were implemented and compared, including Logistic Regression, Random Forest, k-Nearest Neighbors, and XGBoost, in order to evaluate their effectiveness in predicting student dropout.

The results indicate that models based on ensemble and boosting techniques achieve superior performance compared to traditional statistical models. Behavioral features related to students' interaction with learning platforms were found to be among the most influential predictors of dropout risk. The findings demonstrate the feasibility of developing reliable early warning systems using existing educational data. This research shows how predictive analytics can help educational institutions make better decisions. It also explains how these models can be used in low-resource environments to help students stay in school and reduce dropout rates.

KEYWORDS

Student Dropout Prediction, Machine Learning, Early Warning Systems, Learning Analytics, XGBoost, OULAD Dataset

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In the beginning, I praise and thank **Allah** Almighty for bringing us to this moment that I have long wished for and dreamed of.All praise and gratitude be to Him for the blessings of success, patience, and determination.

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I dedicate this research to the **University of Khartoum, Faculty of Mathematical Sciences**, and to all the professors and administrators who provided us with an inspiring educational environment and continuous support throughout our academic career.

Contents

Abstract **i**

Keywords **ii**

Aknowledgement **iii**

List of Figures **vii**

List of Tables **viii**

1 Introduction **1**

1.1 Background 1

1.2 Problem Statement 2

1.3 Motivation 2

1.4 Research Questions 3

1.5 Scope of the Study 3

2 Literature Review **4**

2.1 Predictive Model For Student Dropout 5

2.2 Factor Affecting Student Dropout 5

2.3 Data Sources, Types, and Collection Methods 5

3 Methodology **8**

3.1 Data description 8

3.2 Data Component 8

3.3 Data preprocessing 9

3.4 Research Design and Predictive Modeling 10
3.5 Model Evaluation 11

4 Results **12**
4.1 Phase One: Analysis Of One Course Dropout Dataset 12
4.1.1 Overview of Dropout Distribution 12
4.1.2 Descriptive Statistics and Feature Analysis 13
4.1.3 Feature Importance 13
4.1.4 Model Performance Evaluation 16
4.2 Phase Two:Analysis Of Two Courses Dropout Dataset 20
4.2.1 Overview of Dropout Distribution 20
4.2.2 Descriptive Statistics and Feature Analysis 21
4.2.3 Feature Importance 21
4.2.4 Model Performance 23
4.3 Phase Three: Analysis Of Three Courses Dropout Dataset 26
4.3.1 Overview and Research Limitations 26
4.3.2 Descriptive Statistics 27

5 Discussion **28**
5.1 Factors Associated with Student Dropout 28
5.2 Development of a Predictive Early Warning Model 28
5.3 Comparative Analysis of Predictive Models 29
5.4 Practical Implications for Educational Institutions 29
5.5 Applicability in Low-Resource Educational Environments 30
5.6 Study Limitations 30

6 Conclusion and Recommendations **31**
6.1 Conclusion 31
6.2 Recommendations 31
6.3 Future Work 32

Bibliography **33**

List of Figures

4.1.1 Dropout Distribution for one course drop out 12
4.1.2 Demographic Feature Comparison for one course drop out 13
4.1.3 Normality Testing Results for one course drop out 14
4.1.4 Mann–Whitney U Test: Significant Features for one course drop out 15
4.1.5 Cramer's V Analysis for one course drop out 16
4.1.6 Data Partitioning Strategy (80/20 Split) 16
4.1.7 Logistic Regression Confusion Matrix 17
4.1.8 Logistic Regression Classification 17
4.1.9 Random Forest Confusion Metrics 17
4.1.10 Random Forest 18
4.1.11 KNN Confusion Metrics 18
4.1.12 KNN 19
4.1.13 XGBoost Confusion Metrics 19
4.1.14 XGBoost Performance Metrics 20
4.2.1 Dropout Distribution for Two courses drop out 20
4.2.2 Dropout Distribution for Two courses drop out 21
4.2.3 Shapiro-Wilk Test Result for Two courses drop out 22
4.2.4 Mann–Whitney U Test: Significant Features for one course drop out 22
4.2.5 Data Partitioning Strategy (80/20 Split) 23
4.2.6 Logistic Regression Evaluation 24
4.2.7 Random Forest Model Evaluation 25
4.2.8 KNN Model Evaluation 25
4.2.9 XGBoost Model Evaluation 26

4.3.1 Dropout Distribution for three Courses drop out 26
4.3.2 Demographic Profile for three course drop out 27

List of Tables

3.4.1 Summary of Predictive Models and References 10

Chapter 1

Introduction

1.1 Background

Education is vital for a nation's growth. It helps people live better and also improves the community. Student dropout, which occurs when students leave school before completing their education, is consider one of the challenges in the education system. People lose skills and knowledge and as a consequence, society and the economy are affected.

The problem is worse in places with limited resources. Many educational institutions do not have enought support for students who want to drop out of Universities. In conflict areas like Sudan the economy is hard. Many universities closed. Many of the students had to move to other places. This made dropout worse.

Technology is getting better now. The advancement of student information systems and the introduction of data and AI can help teachers better monitor how students behave, enabling early detection of student dropout. This can help schools provide early help. It can keep students studying.

However, these tools are not used much in underdeveloped countries or war areas like Sudan. We need more studies to know if they really work. We also need to know how educational institutes can use them.

The scale of the student dropout crisis is **significant**, necessitating increased attention from researchers and policymakers alike. If technology is leveraged effectively through predictive modeling, it can serve as a powerful tool to enhance student retention and ensure the completion of their academic studies

Student dropout remains a critical global challenge that **negatively impacts** educational development and economic growth worldwide . According to recent statistics from UNICEF [18], student dropout rates continue to rise in many regions, with approximately **250 million** children and youth remaining out of school globally. Furthermore, the **UNESCO** [**17**] Global Education Monitoring Report highlights that completion rates for secondary education are significantly lower in developing regions, reaching nearly **58%** in some areas. These alarming figures emphasize the urgent need for advanced predictive interventions and early warning systems to identify at-risk students before they exit the educational system

1.2 Problem Statement

Even though there are many new tools using predictive models to reduce student dropout, these methods are still not used much in educational institutes due to few resources, especially in countries that have conflicts like Sudan. Most past studies focused on schools that already have good resources, so there is a lack of real solutions for Universities in difficult situations. This study tries to examine the adaptability and appliction of these ways to use prediction tools to help Universities give early support to students who might drop out.

1.3 Motivation

The motivation for this research stems from the urgent need to address the rising student drop out rates, a phenomenon that significantly **impedes** global educational and economic progress. While advanced educational systems have begun leveraging Artificial Intelligence (AI) to enhance student retention, a critical gap remains in implementing these technologies within low-resource and conflict-affected environments.

In Sudan, the educational crisis has reached a **critical** level due to ongoing conflict and severe economic volatility. Traditional **educational settings** in the region frequently struggle with data fragmentation and the loss of institutional records, making it nearly impossible to identify at-risk students through conventional means. This dire situation serves as a primary driver for this study: the necessity of demonstrating how AI-driven predictive analytics can be adapted to function effectively even amidst infrastructure constraints and limited data availability.

This work is motivated by the potential to provide educators and policymakers with a proactive early warning framework. By transforming existing digital footprints into actionable insights, institutions can execute timely interventions, ensuring that students in vulnerable contexts receive the support needed to successfully complete their academic journeys and contribute to the future recovery of their communities.

1.4 Research Questions

1. What are the main factors that make students at risk of dropping out?
2. How can a trial model be developed using available data to demonstrate the use of early warning systems in supporting at-risk students?
3. How can these models be utilized **within the field of education in general** to reduce dropout and improve student retention
4. How effective are these models in low-resource environments such as Sudan?

1.5 Scope of the Study

The scope of this research focuses on higher education institutions utilizing **Virtual Learning Environments (VLE)**. By analyzing digital footprints from these platforms, the study evaluates the ability of **predictive models** to learn from and identify student interaction patterns

Chapter 2

Literature Review

Among the many open problems in the learning process, students dropout is one of the most complicated and negative ones, both for the student and the institutions, and being able to predict it could help to alleviate its social and economic costs. [5]

Student dropout occurs when a student leaves their program before completing all requirements [7] , [9]. Dropout is a big problem for schools and universities because it affects the success of students, the use of resources, and the reputation of the institution [16] , [1] . It also has social and economic effects, such as reduced job opportunities and waste of human resources [10]

High dropout rates and delayed completion in higher education are associated with considerable personal and social costs. Dropping out from higher education represents a cost for the government and society, an unnecessary expense for the family, and an experience of failure for the university student [2]

students often struggle with abstract thinking and complex problem-solving, leading to higher dropout rates. Similar challenges are seen in other difficult subjects, especially in STEM (Science, Technology, Engineering, and Mathematics) fields like mathematics and engineering. While various factors such as socio-economic conditions and cognitive and metacognitive skills also play a role, the material's high cognitive load and difficulty are significant barriers to successful learning [15] .

Student dropout can be influenced by various factors. Academic performance and engagement with learning materials play a key role [15] , [13] , while mental health issues and food insecurity also contribute significantly to the risk of dropout [19] . Because it is a complex

problem, researchers make predictive models to find students at risk early and help them [11] , [2]

2.1 Predictive Model For Student Dropout

Early studies used traditional statistics such as logistic regression to predict dropout using academic and personal data [7] ,[16] . These models are easy to understand but cannot handle complicated relationships.

Now, studies use machine learning like decision trees, random forests [6],CatBoost , LightGBM, SVM, and neural networks. Research in Nature [15] and Data Mining in Higher Education [1] shows that these models are usually more accurate because they can find complex patterns. in 2023 [14] shows that using data from many sources improves predictions. But some advanced models are hard to explain, which makes teachers or administrators unsure about using them.

2.2 Factor Affecting Student Dropout

Dropout is not only about grades. Students who work while studying are more likely to leave [10] . Teaching quality ,Study [12] indicated that low teacher competence and poor teaching quality can increase student burnout and reduce academic participation, which in turn increases the likelihood of intention to dropout. workload, and academic support also affect students' decisions.

Mental health and social factors matter too. Frontiers in Psychiatry 2024 [19] found that stress increases dropout. Also, digital engagement, such as using online learning platforms, can help keep students if combined with support [13].

2.3 Data Sources, Types, and Collection Methods

The accuracy of predictive models depends on the variety and quality of the data used. The studies show a wide range in data types and collection methods, which directly affects model performance and reliability.

- Academic data: Grades, attendance rates, and the number of withdrawn courses were

used in traditional studies [9] , [1], [2], [2], [7], [16]. These data help identify at-risk students but are limited in explaining non-academic reasons for dropout.

- Demographic and social data: Age, gender, family background, and geographic location proved useful when combined with academic indicators [10], [1],[4].

- Psychological and behavioral data: Indicators such as stress levels, academic satisfaction, and digital participation were collected through surveys or tracking digital behavior on learning platforms [19], [13].

- Multi-source data: Combining academic, social, and psychological records with open or external datasets, as in studies [4], [2], [14], produced the highest prediction accuracy but requires advanced infrastructure to manage data and ensure privacy.

Data collection methods included

- Institutional records: The most common source, but limited in the diversity of variables.

- Surveys and questionnaires: Provided qualitative data but may be biased by self-reporting.

- Digital trace data: Data from student interactions with learning management systems [15] , [13].

- Open datasets: Enabled replicable experiments but often lack detailed contextual information [2].

This comparison indicates that integrating multiple data sources is the best approach to balance comprehensiveness and predictive accuracy, which is adopted in the current study by combining academic, social, and behavioral data from various sources.

While the reviewed literature highlights the effectiveness of machine learning in predicting student dropout, this study aims to address specific gaps by focusing on the following contributions :

This research contributes by investigating the specific ability of models like **XGBoost and Random Forest** to extract and learn from complex behavioral patterns within **Virtual**

Learning Environments (VLE). Unlike traditional studies that rely on static records, this work focuses on dynamic interaction data to identify early warning signs of dropout.

Practical Framework for Local Integration , The study provides a validated predictive framework designed to serve as a foundation for modernizing student retention strategies in Sudanese higher education.

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Chapter 3

Methodology

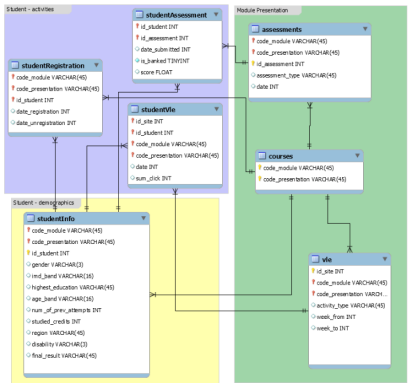
3.1 Data description

We could not find ready-to-use educational data from Sudanese schools or universities. So, we used a public dataset from the internet called the Open University Learning Analytics Dataset (OULAD). It comes from a British open university and has information about students, courses, and how students use the Virtual Learning Environment (VLE).

This data set is open to all and has been used in studies on learning analytics and predicting student dropout. We chose it to show how predictive models can study dropout risks. Later, similar local data could be used for Sudanese institutions.

3.2 Data Component

Data description
Database schema



- The data set has several CSV files, each with different information

- courses. csv – information about courses and how long they last.
- assessments. csv – details of tests and exams.
- studentInfo. csv – student details like age, gender, education level, region, and final results.
- studentAssessment. csv – results of students in tests and assignments.
- vle. csv – learning materials on the VLE, such as HTML pages and PDFs.
- studentVle. csv – shows how students interact with the materials and how many times.
- studentRegistration. csv – registration and withdrawal dates.

This data set is large and organized. It includes both academic and behavioral information, making it useful to build models that predict student dropout.

3.3 Data preprocessing

The original OULAD files were not used directly. Instead, several columns from different files were first joined together to create a single unified record containing each student's information, learning interactions, and academic results of each student. After that, an aggregation step was performed to summarize the data for each student into unified values, such as the total number of interactions or the average grades. Through these steps, a main table was produced, which served as the basis for the feature selection stage. All remaining steps for the cleaning, encoding and data-preparation steps for predictive modeling were applied to this table.

This study follows an **applied and predictive research methodology**. Initially, the dataset was divided into three groups based on the number of courses taken by each student (one course, two courses, and three courses). This grouping was performed to examine the effect of academic workload on student dropout behavior.

3.4 Research Design and Predictive Modeling

Student dropout was defined as a student withdrawing from all courses in which they were registered. The dependent variable across all groups was the dropout status (dropout / non-dropout), while the independent variables included demographic characteristics, academic performance indicators, and students' interactions with the Virtual Learning Environment (VLE).

In the feature selection stage, different statistical techniques were applied according to the type of variables. For categorical variables, the **Chi-square test** was used to assess their association with the target variable. For numerical variables, the **Shapiro–Wilk test** was first applied to examine normality. Since most numerical variables were found to be non-normally distributed, the **Mann–Whitney test** was used to identify statistically significant features.

After feature selection, the data for each group were split into **training and testing sets using an 80/20 ratio**, with 80% of the data used for model training and 20% reserved for testing.

Based on findings from the literature review, **boosting models** were selected as the primary predictive models due to their strong performance in dropout prediction tasks. In addition, several baseline models were implemented for performance comparison, including **Logistic Regression , Random Forest, and k-Nearest Neighbors (k-NN)**.

Table 3.4.1: Summary of Predictive Models and References

Model Name	**Key Concept**	**Source**
Logistic Regression	Estimates dropout probability using a logistic function for binary classification.	[8]
Random Forest	An ensemble of decision trees to enhance stability and reduce overfitting.	[8]
XGBoost	A scalable gradient boosting system known for high performance.	[3]
k-NN	Classifies based on the proximity to the nearest students in the dataset.	[8]

3.5 Model Evaluation

Model performance was evaluated using the **Precision , Recall , F1-score and ROC-AUC** metrics. Given the class imbalance in the dataset, particular emphasis was placed on the **weighted average** of these metrics to ensure a more reliable assessment of model performance.

Chapter 4

Results

In this section, we present the results of our analysis, structured as follows

4.1 Phase One: Analysis Of One Course Dropout Dataset

4.1.1 Overview of Dropout Distribution

The initial phase of this study focuses on students registered for only one course. Figure 4.1.1 illustrates the distribution of dropout versus non-dropout instances, providing a baseline for the subsequent predictive analysis.

```
--- Student Dropout Statistics ---
Total Number of Students: 26306
-----------------------------------
Status: Not Dropout
Count:  19087 students
Ratio:  72.56%
--------------------
Status: Dropout
Count:  7219 students
Ratio:  27.44%
--------------------
```

Figure 4.1.1: Dropout Distribution for one course drop out

4.1.2 Descriptive Statistics and Feature Analysis

Demographic analysis, as shown in Figure 4.1.2, suggests that variables such as gender, region, and disability status are relatively balanced between the two classes. This indicates that demographic factors may not be the primary drivers of attrition for this group.

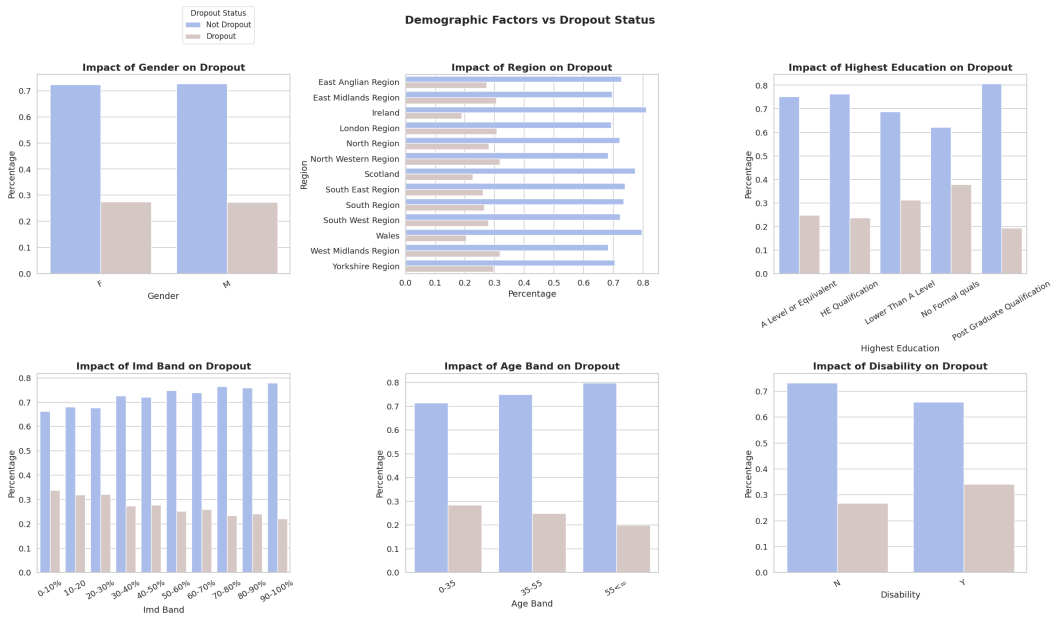


Figure 4.1.2: Demographic Feature Comparison for one course drop out

4.1.3 Feature Importance

To ensure the robustness of our feature selection, a **Shapiro-Wilk test** was applied to check for normality. Given that the results indicated a non-normal distribution ($p < 0.05$), non-parametric evaluations were prioritized.

	Frame	Column	Shapiro_Dropout_p	Shapiro_NotDropout_p	\
1	AAA_2013J_TMA	AAA_2013J_TMA	3.565541e-01	1.876603e-05	
3	AAA_2014J_TMA	AAA_2014J_TMA	1.695206e-02	6.392392e-07	
5	BBB_2013B_CMA	BBB_2013B_CMA	7.721196e-16	2.010711e-51	
7	BBB_2013B_TMA	BBB_2013B_TMA	1.475866e-04	2.595729e-20	
9	BBB_2013J_CMA	BBB_2013J_CMA	2.903782e-17	4.173984e-56	
11	BBB_2013J_TMA	BBB_2013J_TMA	9.073230e-02	4.013641e-18	
13	BBB_2014B_CMA	BBB_2014B_CMA	5.041444e-15	7.249540e-49	
15	BBB_2014B_TMA	BBB_2014B_TMA	2.684699e-03	2.119891e-15	
17	BBB_2014J_TMA	BBB_2014J_TMA	7.640389e-15	3.722691e-36	
19	CCC_2014B_CMA	CCC_2014B_CMA	9.569969e-05	2.569298e-22	
21	CCC_2014B_TMA	CCC_2014B_TMA	8.840350e-03	3.292372e-17	
23	CCC_2014J_CMA	CCC_2014J_CMA	1.938110e-08	6.667112e-22	
25	CCC_2014J_TMA	CCC_2014J_TMA	7.466767e-05	2.795176e-18	
27	DDD_2013B_CMA	DDD_2013B_CMA	4.705295e-03	2.029201e-22	
29	DDD_2013B_TMA	DDD_2013B_TMA	5.664040e-03	3.954909e-22	
31	DDD_2013J_TMA	DDD_2013J_TMA	1.894984e-03	6.100955e-25	
33	DDD_2014B_TMA	DDD_2014B_TMA	3.069828e-04	8.510451e-20	
35	DDD_2014J_TMA	DDD_2014J_TMA	7.827812e-05	5.779823e-21	
37	EEE_2013J_TMA	EEE_2013J_TMA	2.605835e-04	1.667645e-17	
39	EEE_2014B_TMA	EEE_2014B_TMA	7.070523e-04	2.769423e-13	
41	EEE_2014J_TMA	EEE_2014J_TMA	4.923832e-03	5.039004e-20	
43	FFF_2013B_CMA	FFF_2013B_CMA	4.963397e-03	6.268705e-26	
45	FFF_2013B_TMA	FFF_2013B_TMA	7.406792e-10	1.077201e-25	
47	FFF_2013J_CMA	FFF_2013J_CMA	3.848166e-02	9.950895e-24	
49	FFF_2013J_TMA	FFF_2013J_TMA	2.341519e-07	5.860952e-28	
51	FFF_2014B_CMA	FFF_2014B_CMA	6.679117e-04	3.946074e-16	
53	FFF_2014B_TMA	FFF_2014B_TMA	4.146543e-10	2.879979e-20	
55	FFF_2014J_CMA	FFF_2014J_CMA	1.512900e-05	1.315571e-21	
57	FFF_2014J_TMA	FFF_2014J_TMA	5.735602e-05	1.680832e-21	
59	GGG_2013J_CMA	GGG_2013J_CMA	2.738198e-05	1.343721e-43	
61	GGG_2013J_TMA	GGG_2013J_TMA	6.582165e-02	8.147563e-11	
63	GGG_2014B_CMA	GGG_2014B_CMA	2.483990e-06	3.470034e-40	
65	GGG_2014B_TMA	GGG_2014B_TMA	3.446764e-01	6.319817e-15	
67	GGG_2014J_CMA	GGG_2014J_CMA	8.613669e-02	4.881595e-31	
69	GGG_2014J_TMA	GGG_2014J_TMA	6.980326e-01	3.494959e-06	

	Levene_p	Dropout_Normal?	NotDropout_Normal?	Equal_Variance?
1	8.918021e-03	True	False	False
3	7.324728e-01	False	False	True
5	5.225182e-05	False	False	False
7	4.877751e-01	False	False	True
9	2.873297e-04	False	False	False
11	1.157721e-02	True	False	False
13	2.781482e-04	False	False	False
15	1.668915e-01	False	False	True
17	2.707937e-11	False	False	False
19	1.289383e-08	False	False	False
21	1.298344e-04	False	False	False
23	6.176172e-13	False	False	False
25	2.098224e-03	False	False	False
27	2.804547e-05	False	False	False
29	2.273323e-03	False	False	False
31	7.074472e-05	False	False	False
33	1.800633e-04	False	False	False
35	4.737963e-09	False	False	False
37	3.118049e-02	False	False	False
39	4.492490e-03	False	False	False
41	3.621823e-10	False	False	False
43	3.479258e-03	False	False	False
45	8.636323e-06	False	False	False
47	2.894020e-03	False	False	False
49	2.294795e-03	False	False	False
51	5.079350e-03	False	False	False
53	2.854204e-04	False	False	False
55	2.046200e-05	False	False	False
57	8.073169e-04	False	False	False
59	1.292518e-02	False	False	False
61	8.456647e-01	True	False	True
63	1.261520e-01	False	False	True
65	4.757851e-01	True	False	True
67	2.111771e-04	True	False	False
69	1.006502e-01	True	False	True

Figure 4.1.3: Normality Testing Results for one course drop out

Furthermore, the **Mann-Whitney U test** and **Cramer's V** were utilized to quantify the relationship between features and the target variable (Dropout). As seen in Figures 4.1.4

and 4.1.5, academic engagement features show a stronger correlation with student retention.

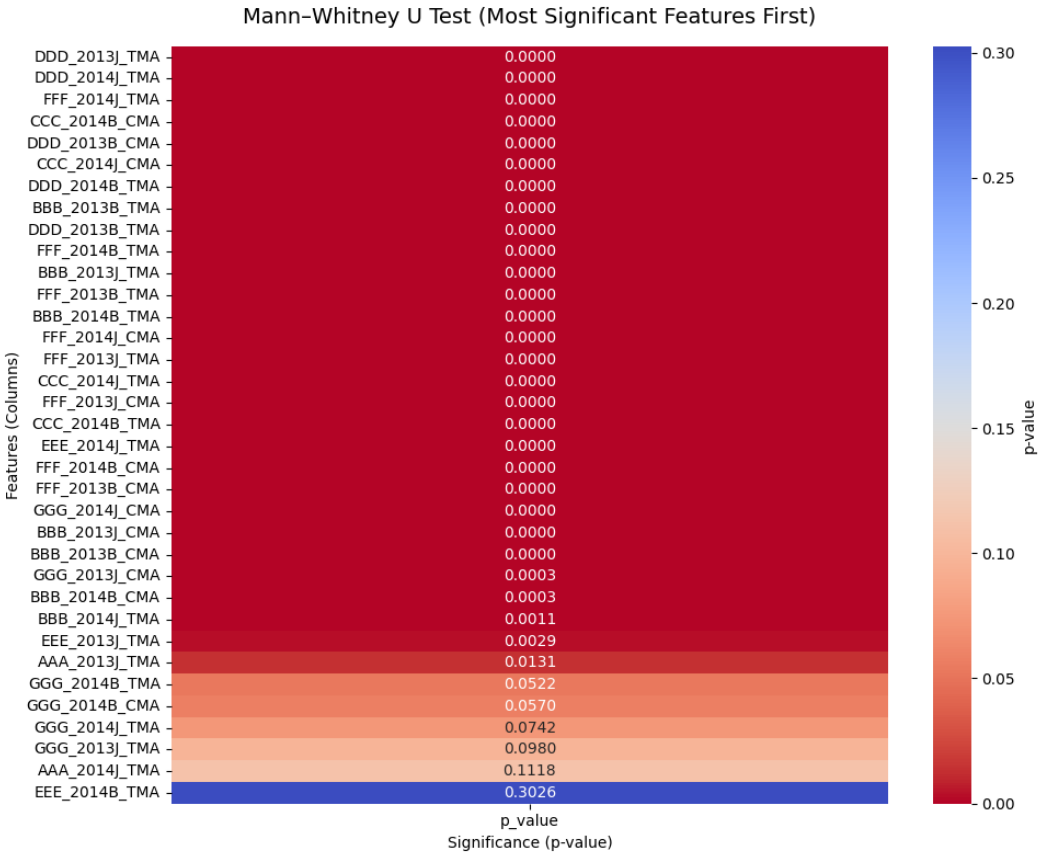


Figure 4.1.4: Mann–Whitney U Test: Significant Features for one course drop out

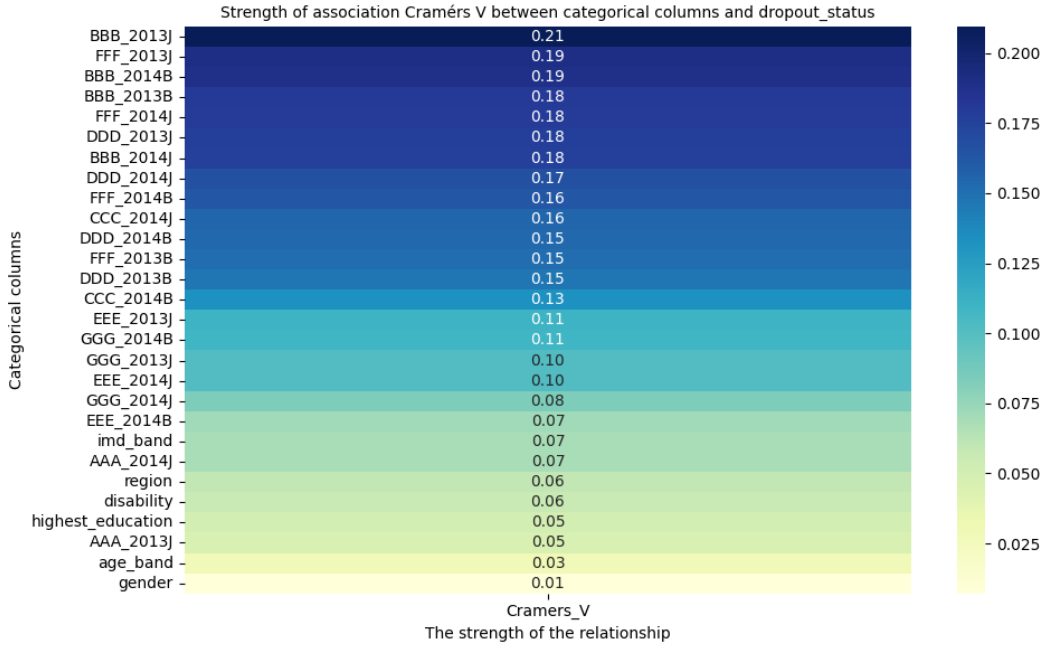


Figure 4.1.5: Cramer's V Analysis for one course drop out

4.1.4 Model Performance Evaluation

The dataset was partitioned into an 80% training set and a 20% testing set. The evaluation metrics for the deployed machine learning models are presented below.

```
Train set:
dropout_status
Not Dropout    3715
Dropout        1316
Name: count, dtype: int64

Test set:
dropout_status
Not Dropout    949
Dropout        309
Name: count, dtype: int64
```

Figure 4.1.6: Data Partitioning Strategy (80/20 Split)

The following figures represent the predictive accuracy for students in Group 2 using Logistic Regression , Random Forest , KNN and XGBoost

```
Classification report for the Logistic Regression model:
              precision    recall  f1-score   support

           0       0.98      1.00      0.99       949
           1       1.00      0.94      0.97       309

    accuracy                           0.99      1258
   macro avg       0.99      0.97      0.98      1258
weighted avg       0.99      0.99      0.99      1258
```

Figure 4.1.7: Logistic Regression Confusion Matrix

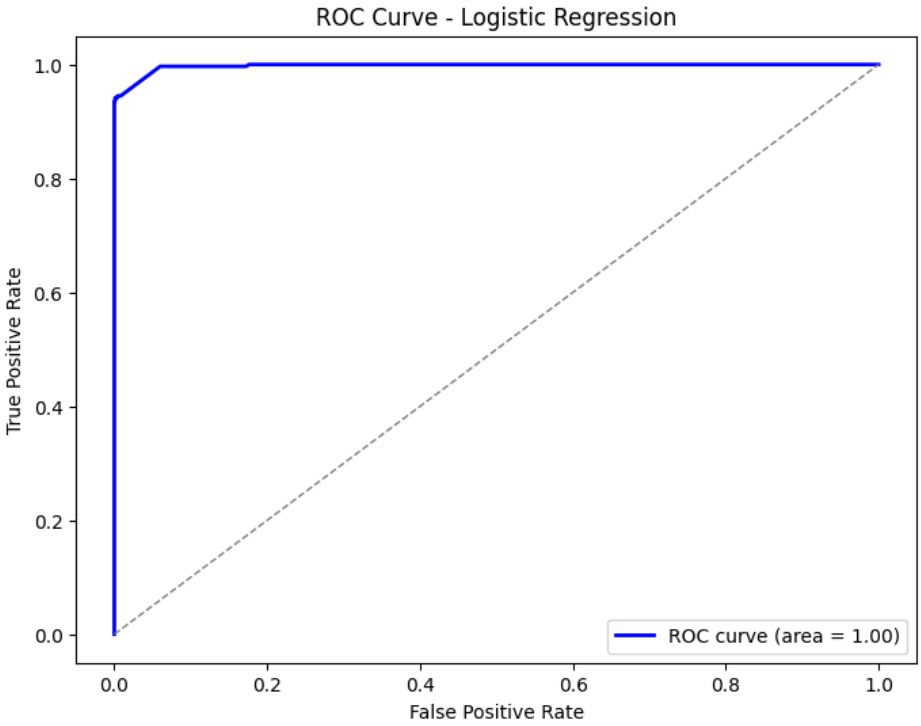


Figure 4.1.8: Logistic Regression Classification

```
Classification report for the random forest model:
              precision    recall  f1-score   support

           0       0.98      1.00      0.99       949
           1       1.00      0.94      0.97       309

    accuracy                           0.99      1258
   macro avg       0.99      0.97      0.98      1258
weighted avg       0.99      0.99      0.99      1258
```

Figure 4.1.9: Random Forest Confusion Metrics

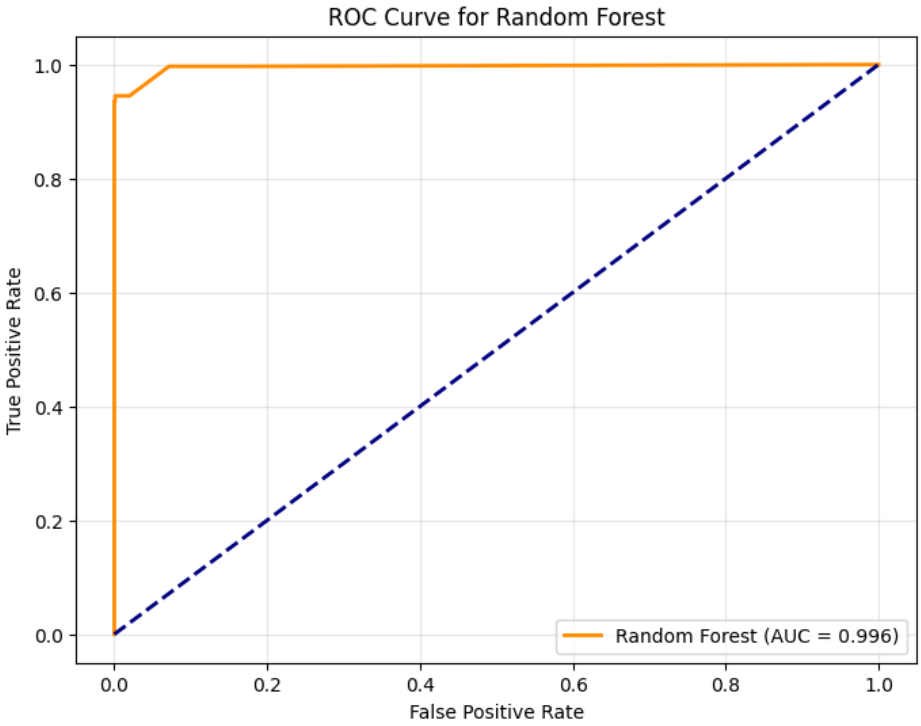


Figure 4.1.10: Random Forest

```
KNN Model Classification Report:
              precision    recall  f1-score   support

           0       0.89      0.99      0.94       949
           1       0.95      0.63      0.76       309

    accuracy                           0.90      1258
   macro avg       0.92      0.81      0.85      1258
weighted avg       0.91      0.90      0.89      1258
```

Figure 4.1.11: KNN Confusion Metrics

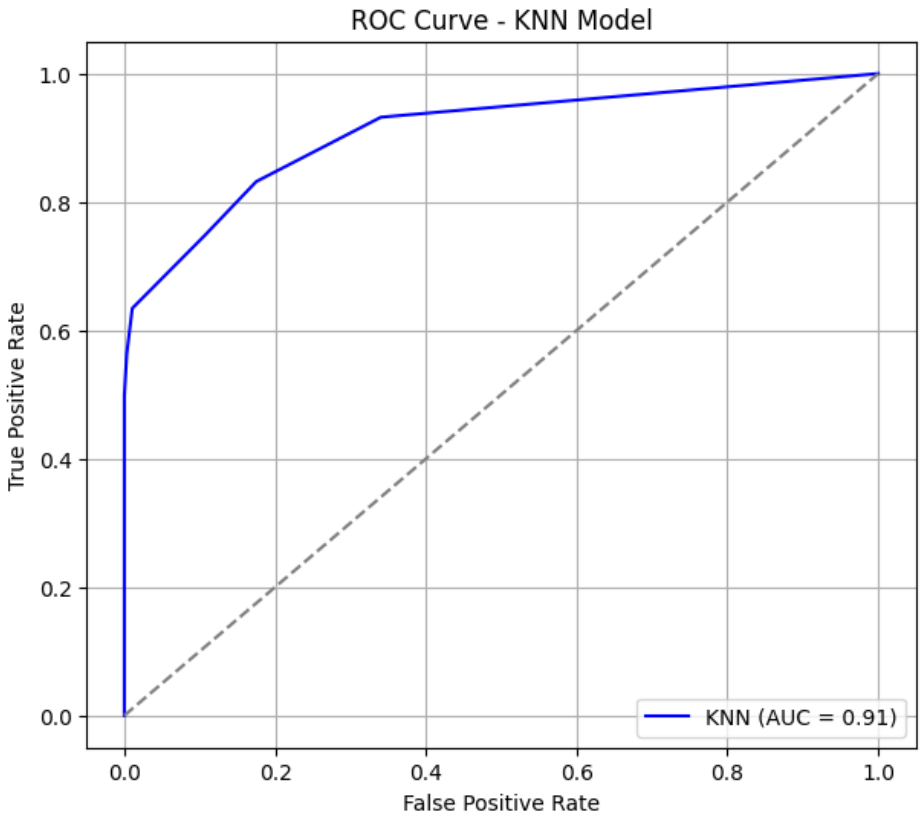


Figure 4.1.12: KNN

```
XGBoost Model Classification Report:
               precision    recall  f1-score   support

 Not Dropout       0.98      1.00      0.99       949
     Dropout       1.00      0.94      0.97       309

    accuracy                           0.98      1258
   macro avg       0.99      0.97      0.98      1258
weighted avg       0.99      0.98      0.98      1258
```

Figure 4.1.13: XGBoost Confusion Metrics

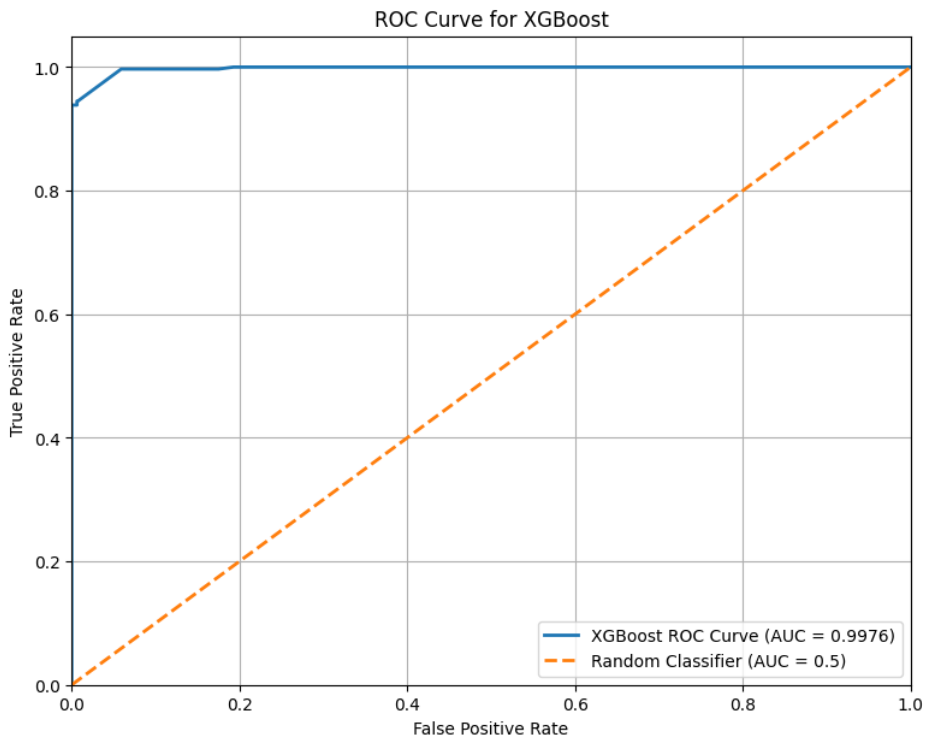


Figure 4.1.14: XGBoost Performance Metrics

4.2 Phase Two:Analysis Of Two Courses Dropout Dataset

4.2.1 Overview of Dropout Distribution

This phase explores students managing a higher academic load. The distribution in Figure 4.2.2 highlights the dropout rate for students enrolled in two courses.

```
--- Student Dropout Statistics ---
Total Number of Students: 2459
-----------------------------------
Status: Not Dropout
Count:  2016 students
Ratio:  81.98%
--------------------
Status: Dropout
Count:  443 students
Ratio:  18.02%
--------------------
```

Figure 4.2.1: Dropout Distribution for Two courses drop out

4.2.2 Descriptive Statistics and Feature Analysis

Demographic analysis, as shown in Figure **??**, suggests that variables such as gender, region, and disability status are relatively balanced between the two classes. This indicates that demographic factors may not be the primary drivers of attrition for this group.

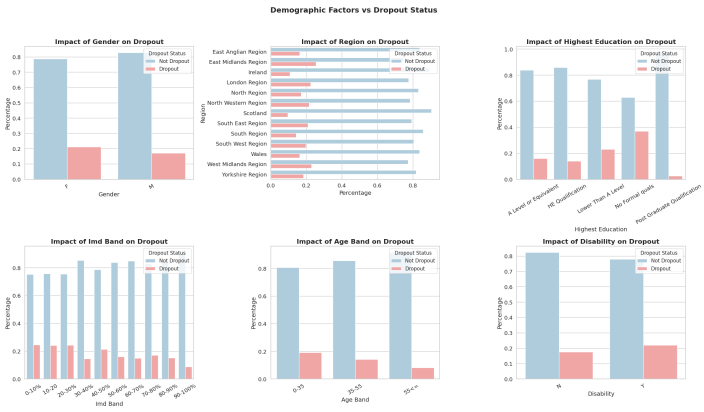


Figure 4.2.2: Dropout Distribution for Two courses drop out

4.2.3 Feature Importance

Consistency in feature importance was maintained by repeating the statistical tests for this cohort. Figure 4.2.4 displays the features that significantly impact dropout prediction in a multi-course context.

```
...              Frame         Column  Shapiro_Dropout_p  Shapiro_NotDropout_p  \
1   CCC_2014B_CMA  CCC_2014B_CMA           0.087756          1.517502e-19
3   CCC_2014B_TMA  CCC_2014B_TMA           0.075868          5.827007e-20
5   CCC_2014J_CMA  CCC_2014J_CMA           0.000059          8.711324e-30
7   CCC_2014J_TMA  CCC_2014J_TMA           0.002699          6.719275e-26
9   DDD_2013B_CMA  DDD_2013B_CMA           0.309323          2.984107e-12
11  DDD_2013B_TMA  DDD_2013B_TMA           0.164987          1.279169e-09
13  DDD_2013J_TMA  DDD_2013J_TMA           0.077173          1.821018e-18
15  DDD_2014B_TMA  DDD_2014B_TMA           0.667005          1.094432e-08
17  DDD_2014J_TMA  DDD_2014J_TMA           0.056161          2.402453e-07
19  EEE_2013J_TMA  EEE_2013J_TMA           0.424652          5.730241e-20
21  EEE_2014B_TMA  EEE_2014B_TMA           0.017520          2.267067e-18
23  EEE_2014J_TMA  EEE_2014J_TMA           0.005990          4.991988e-15
25  FFF_2013J_CMA  FFF_2013J_CMA           0.056394          1.538738e-11
27  FFF_2013J_TMA  FFF_2013J_TMA           0.044322          3.053064e-09
29  FFF_2014B_CMA  FFF_2014B_CMA           0.097450          4.393332e-09
31  FFF_2014B_TMA  FFF_2014B_TMA           0.208523          6.350510e-11
33  FFF_2014J_CMA  FFF_2014J_CMA           0.217588          3.791669e-08
35  FFF_2014J_TMA  FFF_2014J_TMA           0.070032          6.240754e-07

    Levene_p  Dropout_Normal?  NotDropout_Normal?  Equal_Variance?
1   0.521125            True               False             True
3   0.622290            True               False             True
5   0.004693           False               False            False
7   0.114905           False               False             True
9   0.064000            True               False             True
11  0.737475            True               False             True
13  0.010695            True               False            False
15  0.004858            True               False            False
17  0.218777            True               False             True
19  0.164510            True               False             True
21  0.000768           False               False            False
23  0.000003           False               False            False
25  0.673646            True               False             True
27  0.923122           False               False             True
29  0.018590            True               False            False
31  0.956141            True               False             True
33  0.390872            True               False             True
35  0.759811            True               False             True
```

Figure 4.2.3: Shapiro-Wilk Test Result for Two courses drop out

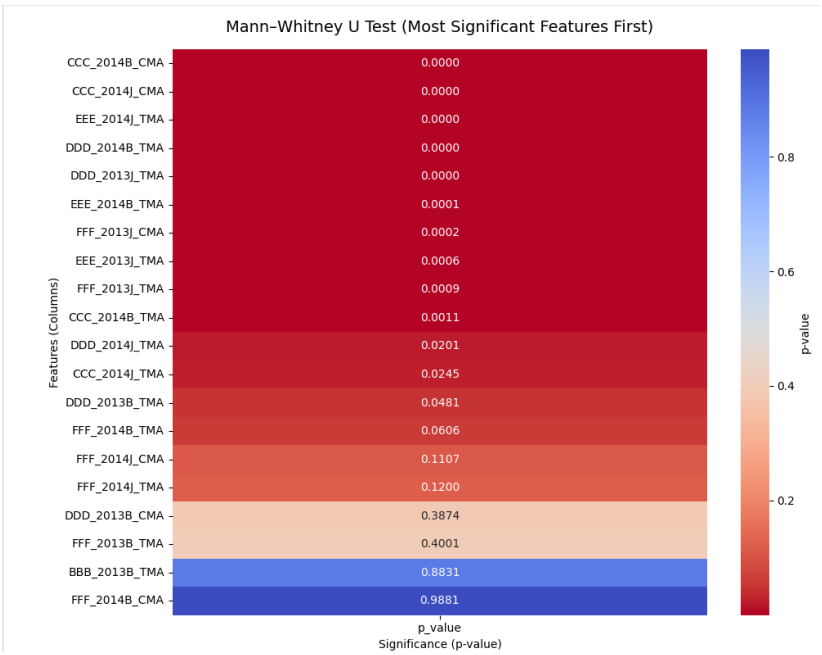


Figure 4.2.4: Mann–Whitney U Test: Significant Features for one course drop out

4.2.4 Model Performance

The dataset was partitioned into an 80% training set and a 20% testing set. The evaluation metrics for the deployed machine learning models are presented below.

```
Train set:
dropout_status
Not Dropout    1614
Dropout         353
Name: count, dtype: int64

Test set:
dropout_status
Not Dropout    402
Dropout         90
Name: count, dtype: int64
```

Figure 4.2.5: Data Partitioning Strategy (80/20 Split)

The following figures represent the predictive accuracy for students in Group 2 using Logistic Regression , Random Forest , KNN and XGBoost.

```
              precision    recall  f1-score   support

           0       0.98      0.99      0.98       402
           1       0.94      0.91      0.93        90

    accuracy                           0.97       492
   macro avg       0.96      0.95      0.96       492
weighted avg       0.97      0.97      0.97       492
```

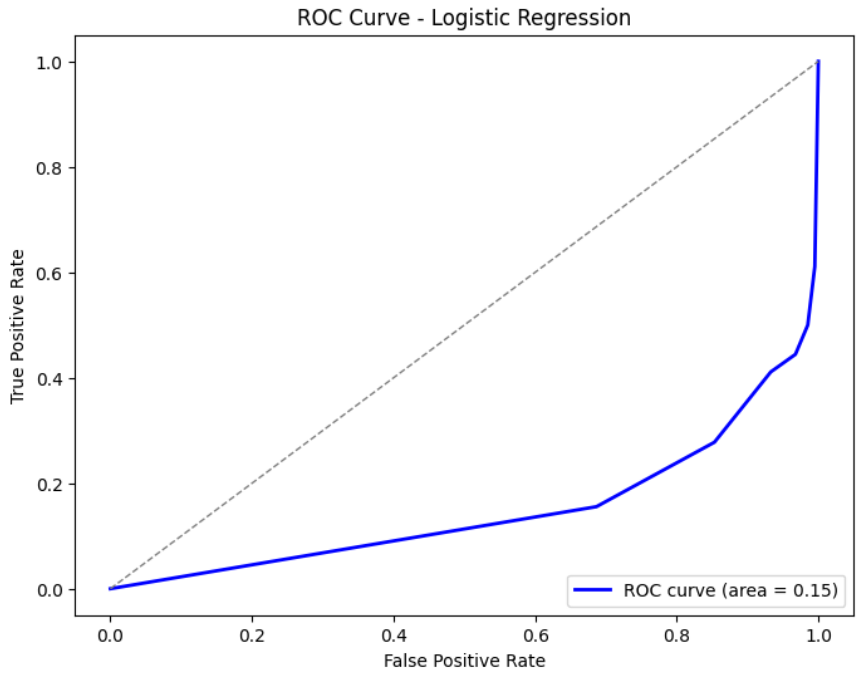


Figure 4.2.6: Logistic Regression Evaluation

```
               precision    recall  f1-score   support

     Dropout       0.95      0.91      0.93        90
 Not Dropout       0.98      0.99      0.99       402

    accuracy                           0.98       492
   macro avg       0.97      0.95      0.96       492
weighted avg       0.98      0.98      0.98       492
```

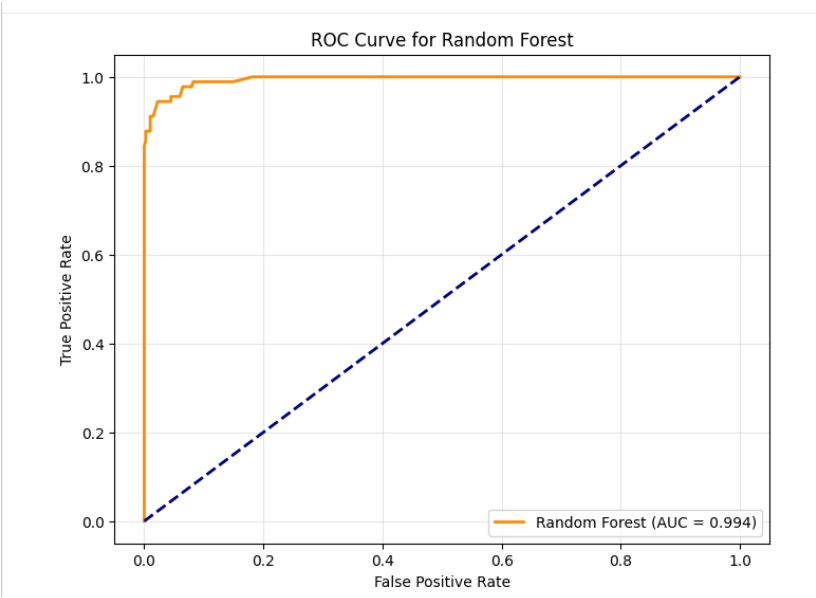


Figure 4.2.7: Random Forest Model Evaluation

	precision	recall	f1-score	support
Dropout	0.66	0.59	0.62	90
Not Dropout	0.91	0.93	0.92	402
accuracy			0.87	492
macro avg	0.79	0.76	0.77	492
weighted avg	0.86	0.87	0.87	492

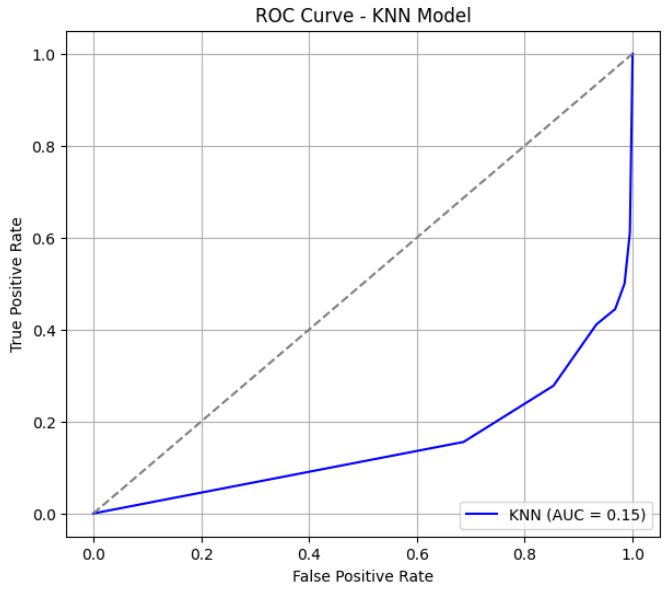


Figure 4.2.8: KNN Model Evaluation

```
              precision    recall  f1-score   support

  Not Dropout       0.98      0.99      0.99       402
      Dropout       0.94      0.92      0.93        90

     accuracy                           0.98       492
    macro avg       0.96      0.95      0.96       492
 weighted avg       0.98      0.98      0.98       492
```

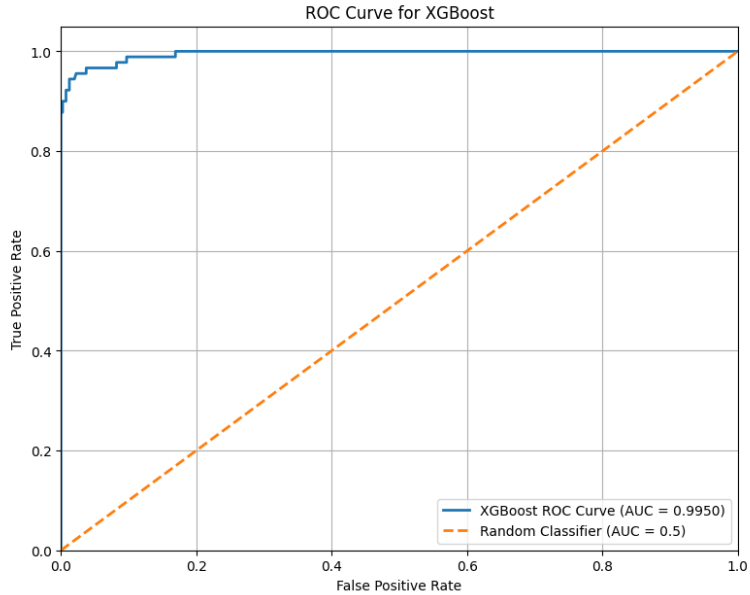


Figure 4.2.9: XGBoost Model Evaluation

4.3 Phase Three: Analysis Of Three Courses Dropout Dataset

4.3.1 Overview and Research Limitations

For students enrolled in three courses, Figure 4.3.1 shows the class distribution.

```
--- Student Dropout Statistics ---
Total Number of Students: 20
-----------------------------------
Status: Not Dropout
Count:  15 students
Ratio:  75.00%
--------------------
Status: Dropout
Count:  5 students
Ratio:  25.00%
--------------------
```

Figure 4.3.1: Dropout Distribution for three Courses drop out

4.3.2 Descriptive Statistics

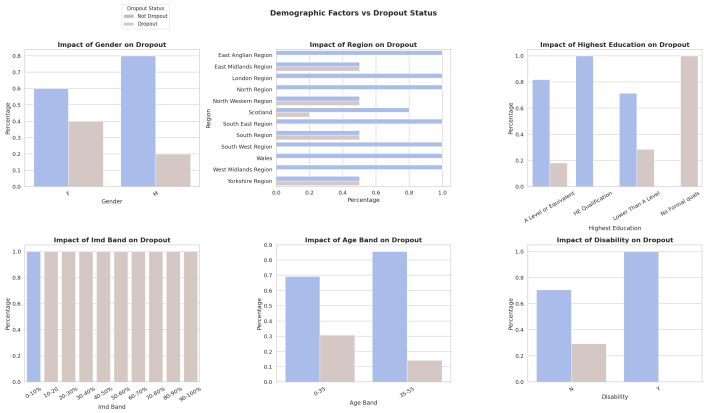


Figure 4.3.2: Demographic Profile for three course drop out

Due to the limited sample size (N=20), this group was excluded from the machine learning training phase to ensure the validity of the study and to avoid the risks of extreme overfitting.

Chapter 5

Discussion

5.1 Factors Associated with Student Dropout

This section discusses the main factors associated with student dropout as identified by the predictive models. The results indicate that behavioral features related to students engagement with the Virtual Learning Environment (VLE) played a significant role in predicting dropout risk. Students with lower levels of interaction, fewer learning activities, and irregular engagement patterns were more likely to drop out.

In addition to behavioral factors, certain academic indicators also contributed to dropout prediction. These findings suggest that student dropout is a multifaceted phenomenon influenced by both academic performance and engagement behavior

5.2 Development of a Predictive Early Warning Model

This study demonstrates how a trial predictive model can be developed using available educational data to support early warning systems. By integrating demographic, academic, and behavioral features, the proposed framework enables the identification of students at risk of dropout at an early stage.

The results showed that boosting-based and random forest models achieved the best predictive performance across all student groups, highlighting their effectiveness in capturing complex patterns in the data. In addition, the Random Forest model also demonstrated strong and competitive performance, indicating its capability to handle nonlinear relation-

ships and feature interactions. These findings confirm the feasibility of developing reliable early warning models using existing institutional data

5.3 Comparative Analysis of Predictive Models

The empirical results demonstrate a high predictive capability across most models. Specifically, **Random Forest** and **XGBoost** consistently maintained an **weighted avg of 0.99** for the dropout class in Phase One. This high level of precision (**1.00**) is crucial for educational institutions, as it ensures that interventions are targeted only at students who are truly at risk. However, the relatively lower weighted avg of the **KNN** model (**0.90**) suggests that distance-based algorithms may struggle with the overlapping behavioral patterns found in imbalanced educational datasets

In Phase Two, while the ensemble models maintained their robustness, a slight decline in predictive sensitivity was observed as academic complexity increased. The **XGBoost** model demonstrated an impressive weighted avg of **0.98**, with a high recall of **0.91** for the dropout class. Conversely, the **KNN** model's performance significantly deteriorated, yielding a weighted avg of only **0.87**. The **ROC Curve** for the KNN model, with an **AUC of 0.15**, indicates that distance-based logic becomes increasingly unreliable as students manage multiple courses simultaneously

5.4 Practical Implications for Educational Institutions

The findings of this study provide practical insights for universities seeking to reduce dropout rates and improve student retention. Predictive models can be incorporated into academic monitoring systems to continuously assess student risk levels.

Early identification of at-risk students allows institutions to implement timely interventions, such as academic advising, targeted support programs, and engagement-enhancing strategies. Consequently, these models can serve as valuable decision-support tools for educational administrators

5.5 Applicability in Low-Resource Educational Environments

Although the analysis was conducted using data from a foreign educational context, the results suggest that the proposed methodology is applicable to low-resource environments such as Sudan. The models rely on data that are relatively easy to collect, including course enrollment records and basic interaction logs.

This indicates that, with minimal infrastructure and data availability, educational institutions in low-resource settings can adopt similar predictive approaches to support student retention initiatives

5.6 Study Limitations

Despite the encouraging results, this study has certain limitations. The use of the OULAD dataset may limit the generalizability of the findings to other educational systems. Additionally, the definition of dropout used in this study may not capture all forms of student disengagement.

It is also worth noting that efforts were made to obtain real educational data from Sudanese institutions. This included contacting the faculty registrar and reaching out through public student platforms to request access to relevant data. However, these attempts were not successful, mainly due to data availability and access constraints. As a result, the study relied on a publicly available dataset to demonstrate the proposed methodology.

Chapter 6

Conclusion and Recommendations

6.1 Conclusion

This study aimed to investigate the factors associated with student dropout and to demonstrate how predictive models can be used as early warning systems in educational contexts. An applied and predictive methodology was adopted, using educational data that included demographic, academic, and behavioral information.

The results showed that predictive modeling techniques can effectively identify students at risk of dropout. In particular, boosting-based models achieved the highest predictive performance across different student groups, while the Random Forest model also demonstrated strong and competitive results. These findings highlight the importance of students' engagement and interaction patterns as key indicators of dropout risk.

Overall, the study confirms the feasibility of developing reliable early warning systems using existing educational data, even in settings with limited resources.

6.2 Recommendations

Based on the findings of this study, the following recommendations are proposed:

- Educational institutions should consider adopting predictive early warning systems to identify at-risk students at an early stage.
- Student engagement data, especially interaction with Virtual Learning Environments

(VLEs), should be continuously monitored and integrated into academic support systems.

- Targeted academic and counseling interventions should be provided to students identified as high risk in order to reduce dropout rates and improve retention.
- Institutions in low-resource environments can benefit from lightweight predictive models that rely on readily available data.
- Although demographic variables showed a limited impact on dropout prediction in the analyzed dataset, this result is likely influenced by the socioeconomic context of the educational system from which the data were obtained. Therefore, when applying the proposed framework in low-resource settings such as Sudan, greater emphasis should be placed on demographic factors, as economic and social conditions may significantly influence students' persistence and dropout behavior

6.3 Future Work

Future research can extend this study by applying the proposed framework to local datasets from Sudanese schools and universities, which would enhance the generalizability and contextual relevance of the findings.

In addition, greater use can be made of university Learning Management Systems (LMS) to collect more comprehensive and detailed data on students' academic activities and engagement behaviors. The availability of such data would support the development of more effective predictive models and enable the application of machine learning and artificial intelligence techniques for early intervention aimed at reducing student dropout.

Furthermore, future studies may incorporate temporal and sequential data to gain deeper insights into students' learning trajectories. Exploring advanced machine learning methods and explainable artificial intelligence approaches could also improve prediction accuracy while enhancing the transparency and interpretability of the proposed model

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